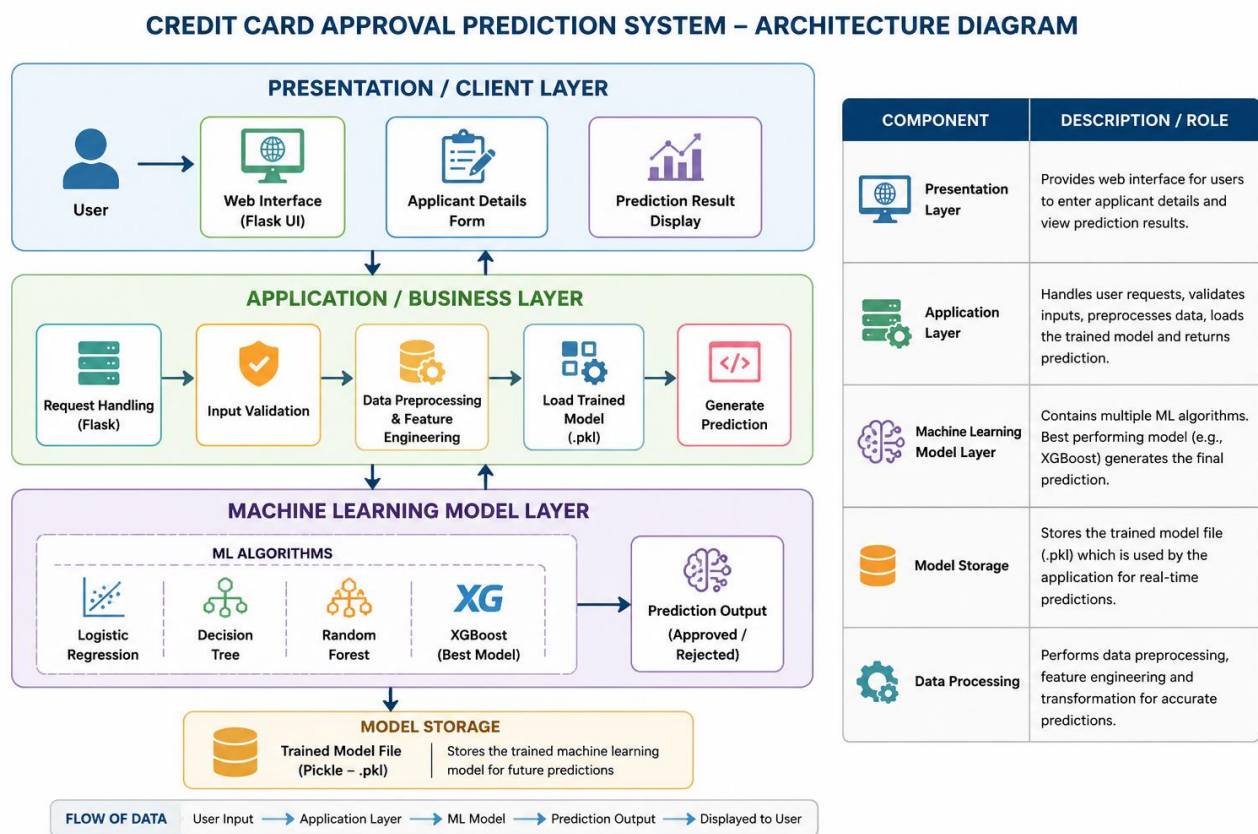


## Solution Architecture

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 1 July 2026                     |
| <b>Team ID</b>       | XXXXXX                          |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Solution Architecture Diagram:



## Component Description Table

| Component Name         | Description / Role in Architecture                                                                              | Technologies Used             |
|------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------|
| Presentation Layer     | Provides the user interface for entering applicant information and viewing prediction results.                  | HTML, JavaScript              |
| Application Layer      | Handles user requests, validates inputs, preprocesses data, loads the trained model, and generates predictions. | Python, Flask                 |
| Machine Learning Layer | Executes the trained classification model to predict whether an applicant is likely to be accepted or rejected. | Scikit-Learn, XGBoost, Pickle |
| Data Processing        | Cleans, transforms, and encodes applicant data before prediction.                                               | Pandas, NumPy                 |